

CS & IT ENGINEERING



Computer Network

Transport Layer

Lecture No. - 09

By - Abhishek Sir





Recap of Previous Lecture



Topic

TCP Flow Control



Topics to be Covered



Topic

TCP Congestion Control

ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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#Q. On a TCP connection, current congestion window size is Congestion Window = 4 KB. The window size advertised by the receiver is Advertise Window = 6 KB. The last byte sent by the sender is LastByteSent = 10240 and the last byte acknowledged by the receiver is LastByteAcked = 8192. The current window size at the sender is _____.

$$CWND = 4KB, RWND = 6KB \quad [GATE-2005]$$

(A) 2048 bytes

$$LastByteAcked = 8192, LastByteSent = 10240$$

✓ (B) 4096 bytes

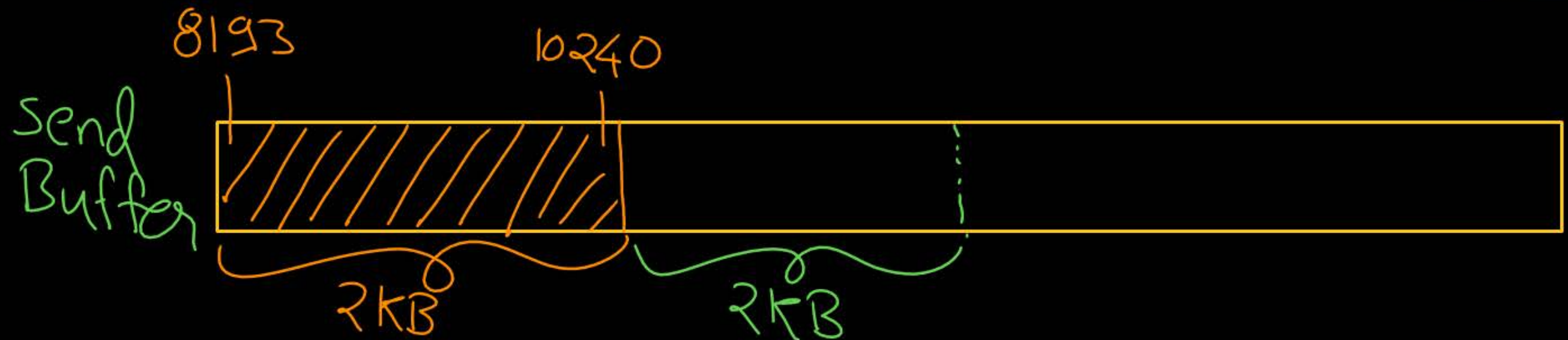
$$[LastByteSent - LastByteAcked] \leq \min[CWND, RWND]$$

$$(10240 - 8192) \leq \min[4KB, 6KB]$$

(C) 6144 bytes

(D) 8192 bytes

Ans: B





Topic : Congestion Control

→ Two phase of TCP congestion control *Algorithm : [AIMD]*

1. Slow Start Phase

[when $CWND < SSThresh$]

2. Congestion Avoidance Phase

[when $CWND \geq SSThresh$]

SSThresh : Slow Start Threshold



Topic : Slow Start Phase



→ When Congestion Window Size is less than Slow Start Threshold

$$\text{CWND} < \text{SSThresh}$$

→ Congestion Window (CWND) grow exponentially

→ CWND increase by 1 MSS on every successful acknowledgment

→ CWND approximately doubles on every round trip time

$$\text{CWND} = \text{CWND} + 1 \text{ MSS}$$

$$\text{CWND} = 1000 \text{ bytes}$$

$$\text{MSS} = 100 \text{ bytes}$$

$$\text{CWND} = 10 \text{ MSS}$$

#Q. In the slow start phase of the TCP congestion control algorithm, the size of the congestion window :

[GATE-2008]

11SC,

- (A) does not increase
- ~~(B)~~ increases linearly
- ~~(C)~~ increases (quadratically)
- ☒ (D) increases exponentially

Ans: D

#Q. Consider the following statements regarding the slow start phase of the TCP congestion control algorithm. Note that cwnd stands for the TCP congestion window and MSS denotes the Maximum Segment Size.

- ☐ i. The cwnd increase by 2 MSS on every successful acknowledgement. FALSE
- ☐ ii. The cwnd approximately doubles on every successful acknowledgement. FALSE
- ☐ iii. The cwnd increase by 1 MSS every round trip time. FALSE
- ☒ iv. The cwnd approximately doubles every round trip time. TRUE

Which one of the following is correct?

[GATE-2018, 1-Mark]
IIT-G

- (A) Only (ii) and (iii) are true
- (B) Only (i) and (iii) are true
- ☒ (C) Only (iv) is true
- (D) Only (i) and (iv) are true

Ans: C



Topic : Congestion Avoidance Phase

→ When Congestion Window Size is greater then equal to Slow Start Threshold

$$\text{CWND} \geq \text{SSThresh}$$

→ Congestion Window (CWND) grow linearly

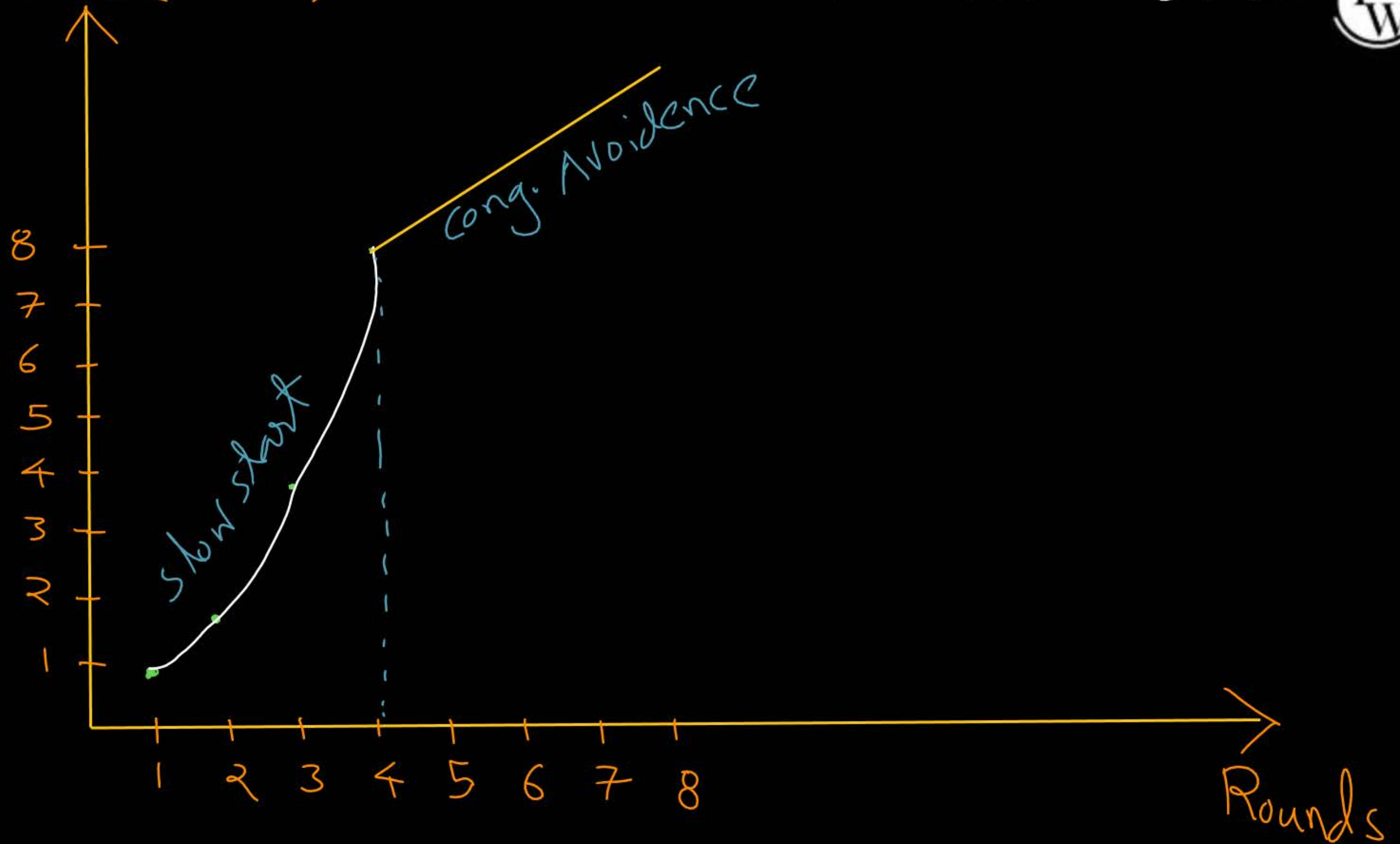
→ CWND increase by 1 MSS on every round trip time
[On successful acknowledgments of all transmitted segments]

Additive Increase

$$\text{CWND} = \text{CWND} + \frac{1}{\text{[CWND]}}$$

CWND (in MSS)

SSThresh = 8 MSS





Topic : Congestion Control

→ When TCP connection established : [During Data Transfer Phase]

Initialize : $SSThresh = (\text{Infinite}) = (2^{16} - 1) * 2^{14} \text{ Bytes}$

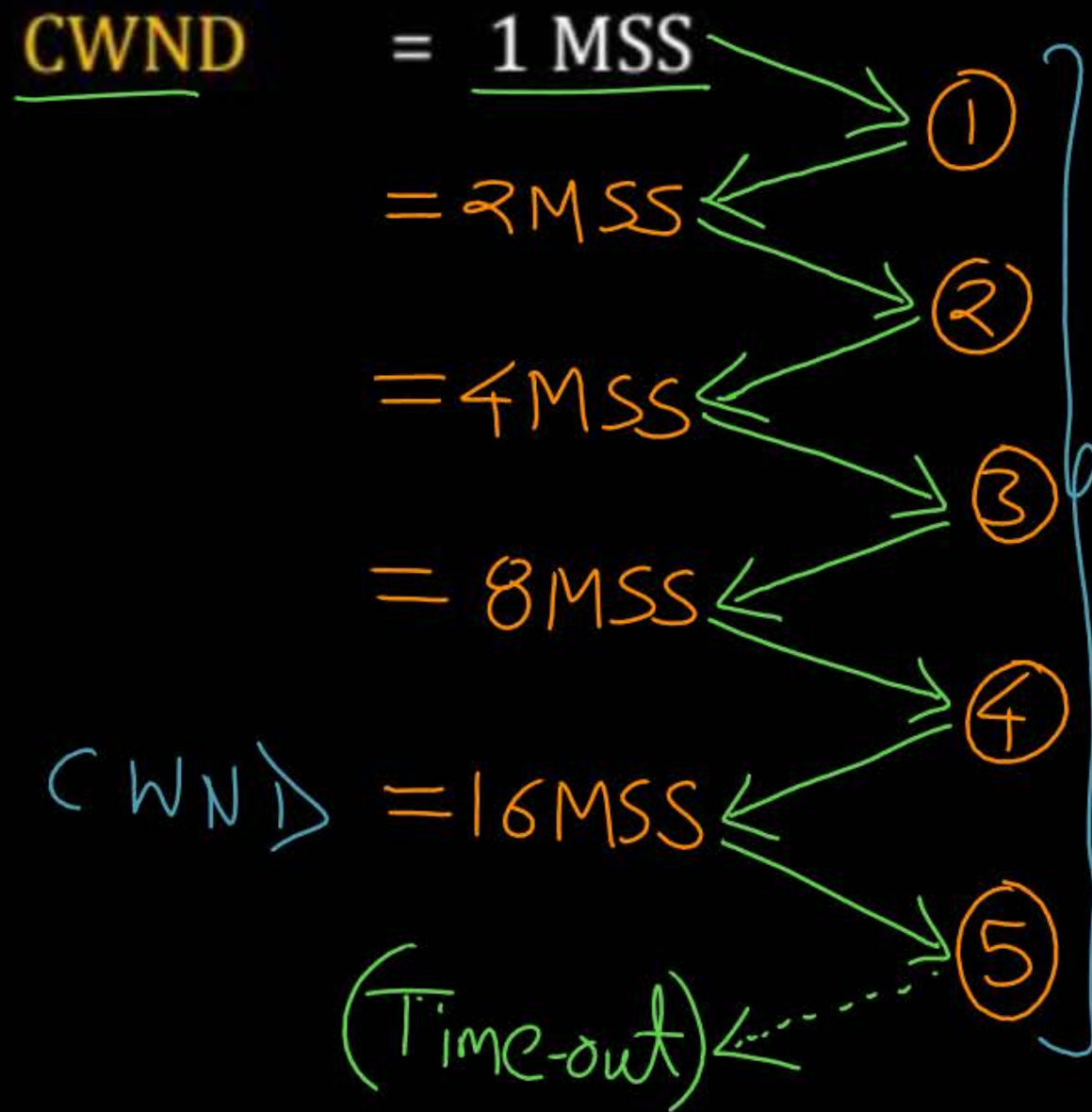
CWND = 1 MSS (Default) / 2MSS / 3MSS / 4MSS

→ When Time-out occur :

Update : $SSThresh = [CWND / 2]$ → Multiplicative Decrease

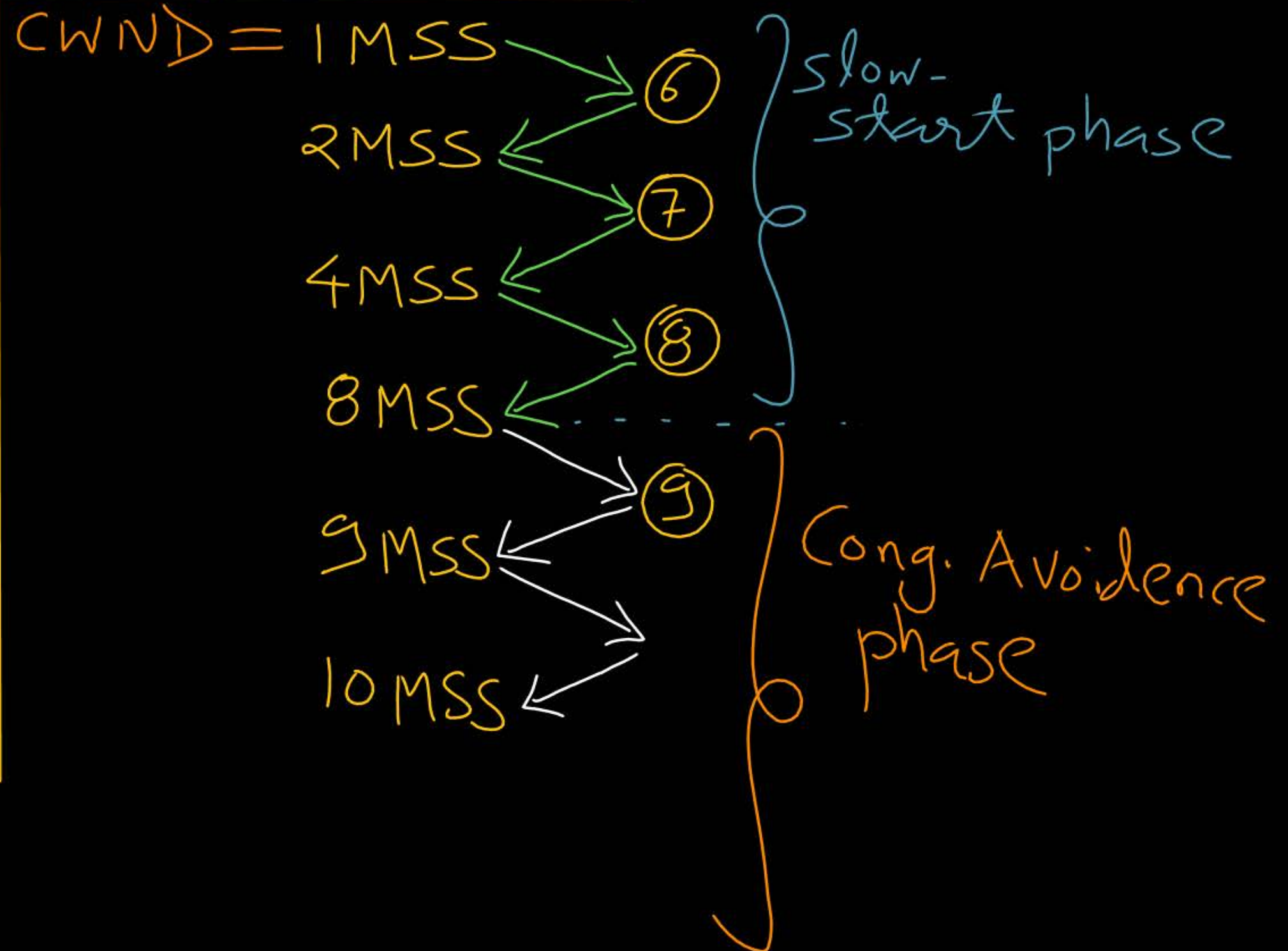
CWND = 1 MSS

SSThresh = Infinite



$$SSThresh = \frac{CWND}{2}$$

$$= \frac{16 MSS}{2} = 8 MSS$$



[NAT]

IISC, H.W.

[GATE-2024][2 Mark]



#Q. Consider a TCP connection operating at a point of time with the congestion window of size 12 MSS (Maximum Segment Size), when a timeout occurs due to packet loss. Assuming that all the segments transmitted in the next two RTTs (Round Trip Time) are acknowledged correctly, the congestion window size (in MSS) during the third RTT will be _____.

#Q. Let the size of congestion window of a TCP connection be 32 KB when a timeout occurs. The round trip time of the connection is 100 msec and the maximum segment size used is 2 KB. The time taken (in msec) by the TCP connection to get back to 32 KB congestion window is _____.

[GATE-2014, Set-1, 2-Mark]

IIT-KGP, H.W.

#Q. Consider a TCP connection between a client and a server with the following specifications; the round trip time is 6 ms, the size of the receiver advertised window is 50 KB, slow-start threshold at the client is 32 KB, and the maximum segment size is 2 KB. The connection is established at time $t=0$. Assume that there are no timeouts and errors during transmission. Then the size of the congestion window (in KB) at time $t+60$ ms after all acknowledgments are processed is _____

[GATE-2020, 2-Mark]

IIT-D, H.W.

#Q. Consider an instance of TCP's Additive Increase Multiplicative Decrease (AIMD) algorithm where the window size at the start of the slow start phase is 2 MSS and the threshold at the start of the first transmission is 8 MSS. Assume that a timeout occurs during the fifth transmission. Find the congestion window size at the end of the tenth transmission.

- (A) 8 MSS
- (B) 14 MSS
- (C) 7 MSS
- (D) 12 MSS

[GATE-2012, 2-Mark]

IIT-D, H.W.



2 mins Summary



Topic

TCP Congestion Control



THANK - YOU

